

DIRECTED ENERGY WEAPONS: High-Energy Laser Weapons



Dr Anil Kumar Maini
is former director,
Laser Science and
Technology Centre,
a premier laser and
optoelectronics
research and
development
laboratory of
Defence Research
and Development
Organisation of
Ministry of Defence

Development of high-energy lasers with military potential is leading to the production of beam weapons that transfer coherent light energy to the intended target to cause structural damage. In the coming years, monopoly of chemically-propelled projectiles may give way to dynamic co-existence and competition with directed energy weapons (DEWs) as the next-generation weapons for both tactical and strategic missions.

Introduction of high-power lasers (HPLs) has enabled execution of many new missions hitherto extremely complex to realise with conventional kinetic energy weapons (KEWs). These include ground based laser for disabling low-Earth orbit satellites, airborne laser for destroying ballistic missiles and space based laser for negation of theatre and inter-continental ballistic missiles. The capability demonstrated by HPLs, particularly in the last 10 to 15 years, to destroy various types of fast-moving aerial targets such as unmanned aerial vehicles has established high-energy laser weapons as the weapons for the 21st century.

Operational advantages

Primary advantages of laser based DEWs include speed-of-light delivery, near-zero collateral damage, multiple target engagement and rapid re-targeting capability, immunity to electromagnetic interference and no influence of gravity. Deep magazine and low cost per shot are the other advantages.

Laser weapons engage targets at speed of light with essentially no time-of-flight as compared to conventional KEWs that require a finite travel time. As an example, one of the world's fastest cruise missiles, BrahMos, with a supersonic speed of 2.8Mach to 3Mach, would take about five minutes to reach its target located at its maximum operational range of 300km.

On the other hand, the same target when

targeted by a laser DEW would be hit in a millisecond. This makes these particularly suitable for engaging fast-moving targets.

Multiple target engagement and rapid retargeting features of laser based DEWs is attributed to their being powered by rechargeable chemical energy (in the case of chemical lasers) or electrical energy (in the case of solid-state and fibre lasers). Shifting from one target to another involves only repointing and refocusing of the beam-directing optical system.

The processes of generation and transfer of lethal laser power to the target are purely in the optical spectrum and, hence, are immune to any electromagnetic interference and jamming. Laser pointing is practically without any inertia, and light bullet has no mass and therefore is not influenced by gravity. As a result, it does not require any mid-course correction.

The total number of shots a laser can fire is limited only by the amount of chemical fuel in the case in chemical lasers or electrical power for solid-state and fibre lasers.

Cost per shot in the case of laser based DEWs is much lower than in the case of conventional KEWs. Projectile weapon systems, guided missiles in particular, expend a lot of expensive hardware such as rocket motors, guidance systems, avionics, seekers, airframes and the like, every time these are fired. In case of laser weapons, cost of each laser firing is essentially the cost of chemical fuel or electrical power consumed.

As an example, a fourth-generation shoulder-fired surface-to-air missile of FIM-92 Stinger series costs about US\$ 40,000. This missile can be used against an aircraft to carry out one such mission. A similar mission from a land based laser DEW system can be carried out with 50kW to 100kW solid-state laser by firing the laser beam for a dwell time of about five seconds. Going by present-day technology level, this laser